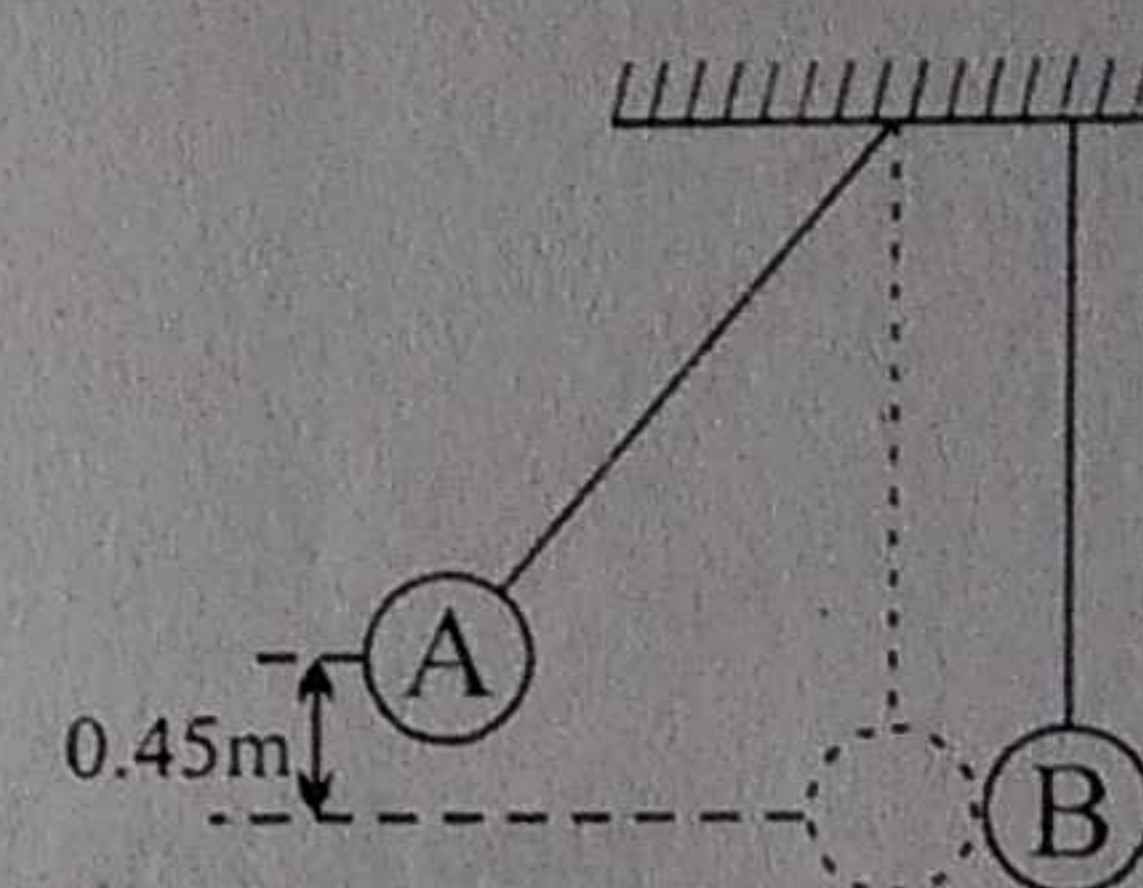
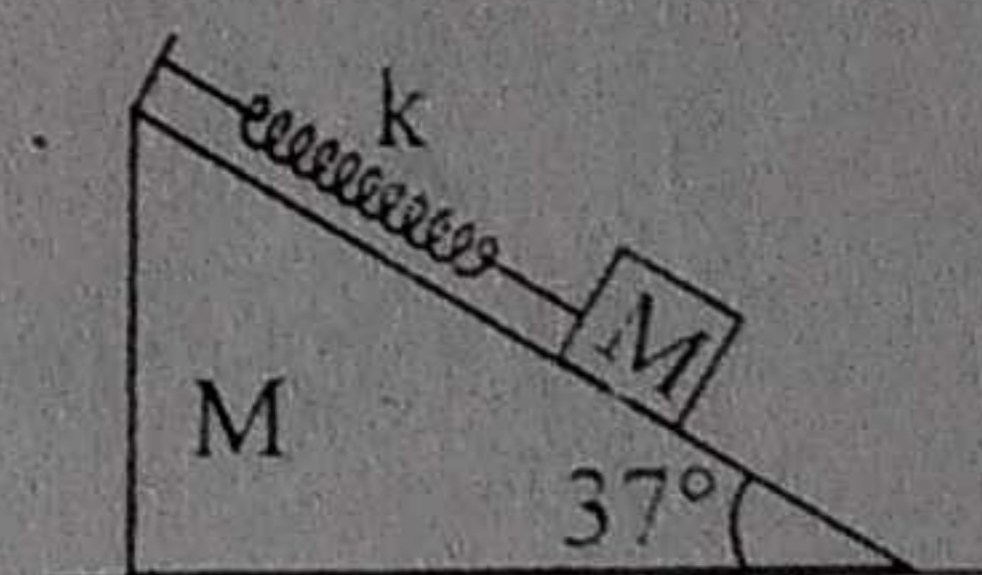
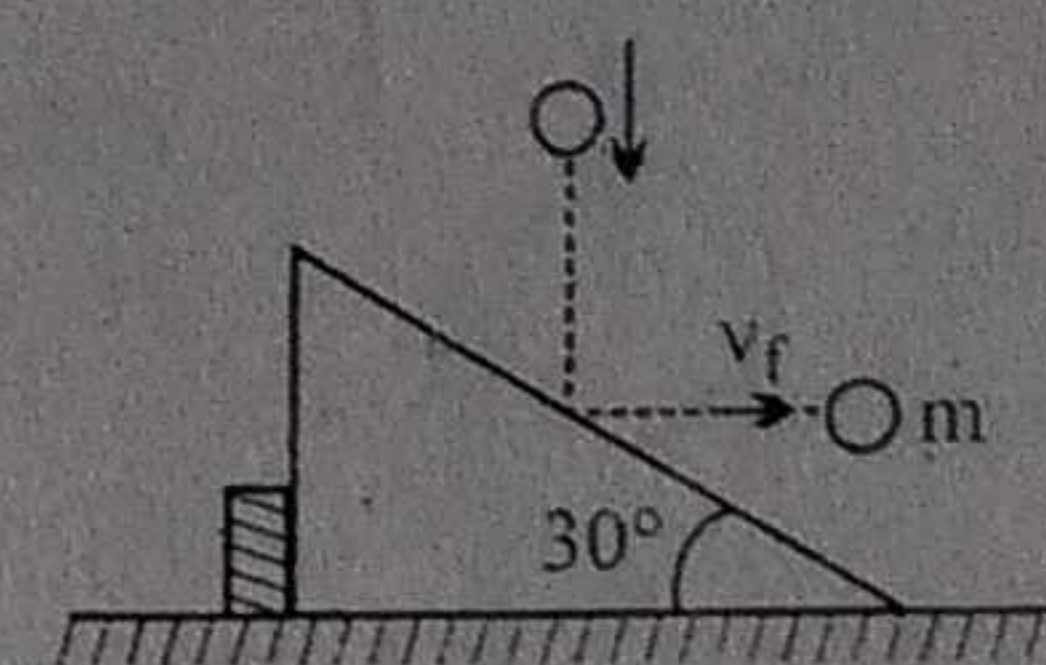


- Q.1 A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What **cannot** be the possible angle of its velocity with the vertical after the collision. (Assume $e \leq 1$).
 (A) 45° (B) 30° (C) 53° (D) 60°
- Q.2 As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30° , the velocity v_f will be
 (A) 3 m/s (B) $\sqrt{3} \text{ m/s}$
 (C) $1/\sqrt{3} \text{ m/s}$ (D) this is not possible
- Q.3 The system of the wedge and the block connected by a massless spring as shown in the figure is released with the spring in its natural length. Friction is absent. maximum elongation in the spring will be
 (A) $\frac{3Mg}{5k}$ (B) $\frac{6Mg}{5k}$ (C) $\frac{4Mg}{5k}$ (D) $\frac{8Mg}{5k}$
- Q.4 Two massless string of length 5 m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to the string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1 J . The velocity of ball A just after the collision is
 (A) 5 ms^{-1} to the right (B) 5 ms^{-1} to the left
 (C) 1 ms^{-1} to the right (D) 1 ms^{-1} to the left
- Q.5 A man of mass M stands at one end of a plank of length L which lies at rest on a frictionless surface. The man walks to other end of the plank. If the mass of the plank is $\frac{M}{3}$, then the distance that the man moves relative to ground is :
 (A) $\frac{3L}{4}$ (B) $\frac{L}{4}$ (C) $\frac{4L}{5}$ (D) $\frac{L}{3}$
- Q.6 Two balls A and B having masses 1 kg and 2 kg , moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then the coefficient of restitution is
 (A) 0.1 (B) 0.2 (C) 0.4 (D) None
- Q.7 A particle of mass $3m$ is projected from the ground at some angle with horizontal. The horizontal range is R . At the highest point of its path it breaks into two pieces m and $2m$. The smaller mass comes to rest and larger mass finally falls at a distance x from the point of projection where x is equal to
 (A) $\frac{3R}{4}$ (B) $\frac{3R}{2}$ (C) $\frac{5R}{4}$ (D) $3R$
- Q.8 A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg . How far is he from the shore at the end of this time ?
 (A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m
- Q.9 From a circle of radius a , an isosceles right angled triangle with the hypotenuse as the diameter of the circle is removed. The distance of the centre of gravity of the remaining position from the centre of the circle is

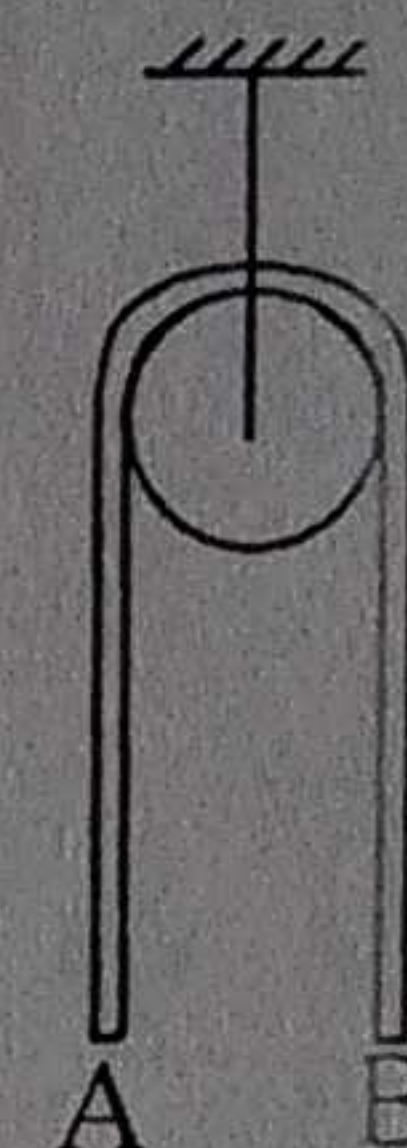


- (A) $3(\pi - 1)a$ (B) $\frac{(\pi - 1)a}{6}$ (C) $\frac{a}{3(\pi - 1)}$ (D) $\frac{a}{3(\pi + 1)}$

- Q.10 A truck moving on horizontal road towards east with velocity 20 ms^{-1} collides elastically with a light ball moving with velocity 25 ms^{-1} along west. The velocity of the ball just after collision
 (A) 65 ms^{-1} towards east (B) 25 ms^{-1} towards west
 (C) 65 ms^{-1} towards west (D) 20 ms^{-1} towards east

Question No. 83 to 84 (2 questions)

A uniform chain of length $2L$ is hanging in equilibrium position, if end B is given a slightly downward displacement the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible



- Q.11 The acceleration of end B when it has been displaced by distance x , is

- (A) $\frac{x}{L}g$ (B) $\frac{2x}{L}g$ (C) $\frac{x}{2}g$ (D) g

- Q.12 The velocity v of the string when it slips out of the pulley (height of pulley from floor $> 2L$)

- (A) $\sqrt{\frac{gL}{2}}$ (B) $\sqrt{2gL}$ (C) $\sqrt{gL}$ (D) none of these

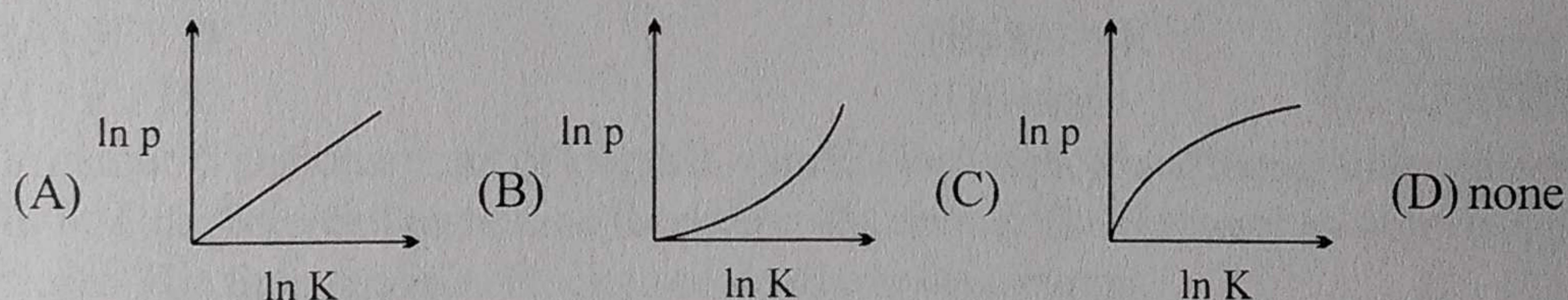
- Q.13 A small sphere is moving at a constant speed in a vertical circle. Below is a list of quantities that could be used to describe some aspect of the motion of the sphere.

- I – kinetic energy
 II – gravitational potential energy
 III – momentum

Which of these quantities will change as this sphere moves around the circle?

- (A) I and II only (B) I and III only (C) III only (D) II and III only

- Q.14 Which of the following graphs represents the graphical relation between momentum (p) and kinetic energy (K) for a body in motion?

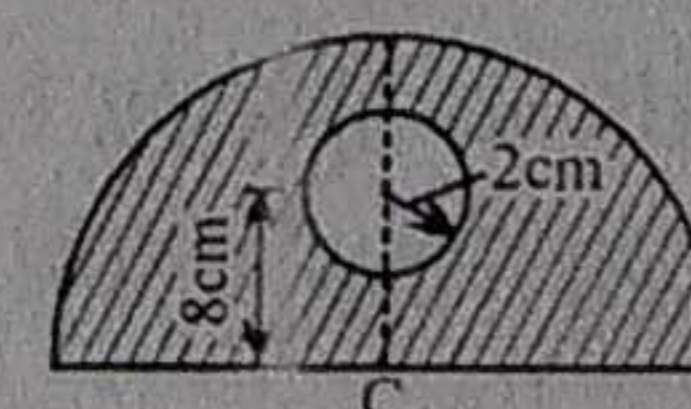


- Q.15 A small bucket of mass M kg is attached to a long inextensible cord of length L m. The bucket is released from rest when the cord is in a horizontal position. At its lowest position, the bucket scoops up m kg of water and swings up to a height h . The height h in meters is

- (A) $\left(\frac{M}{M+m}\right)^2 L$ (B) $\left(\frac{M}{M+m}\right) L$ (C) $\left(\frac{M+m}{M}\right)^2 L$ (D) $\left(\frac{M+m}{M}\right) L$

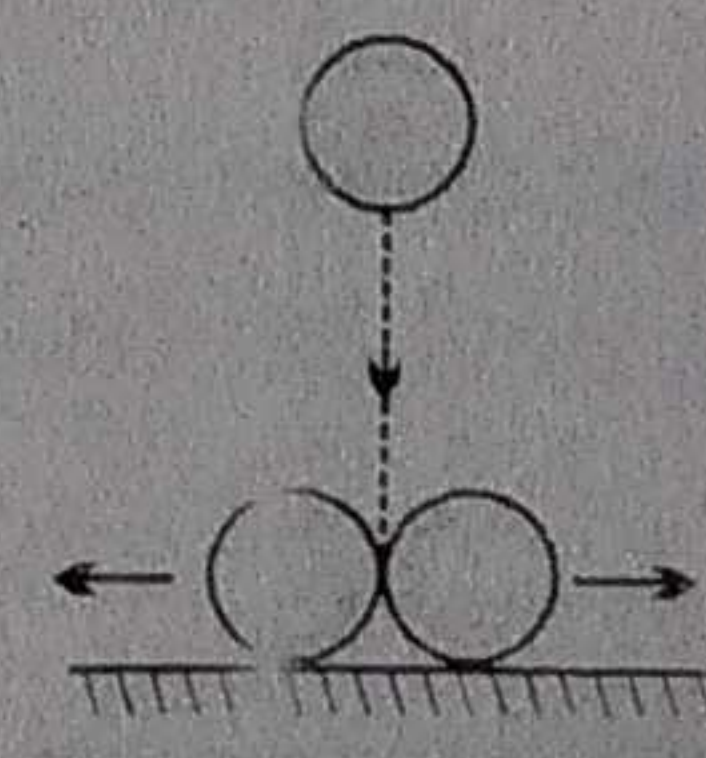
- Q.16 In the figure shown a hole of radius 2 cm is made in a semicircular disc of radius 6 cm at a distance 8 cm from the centre C of the disc. The distance of the centre of mass of this system from point C is :

- (A) 4 cm (B) 8 cm (C) 6 cm (D) 12 cm

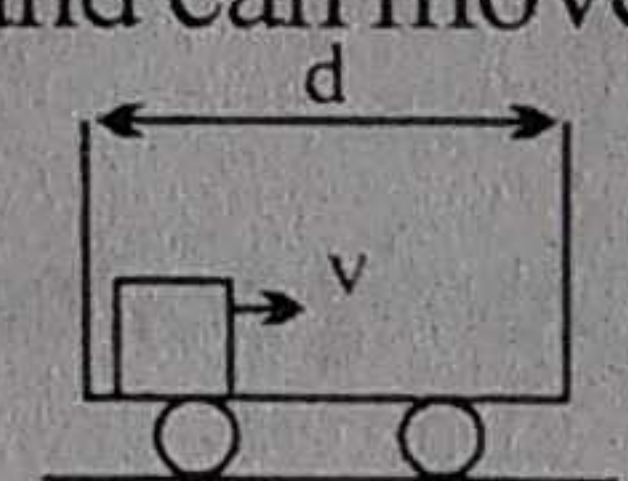
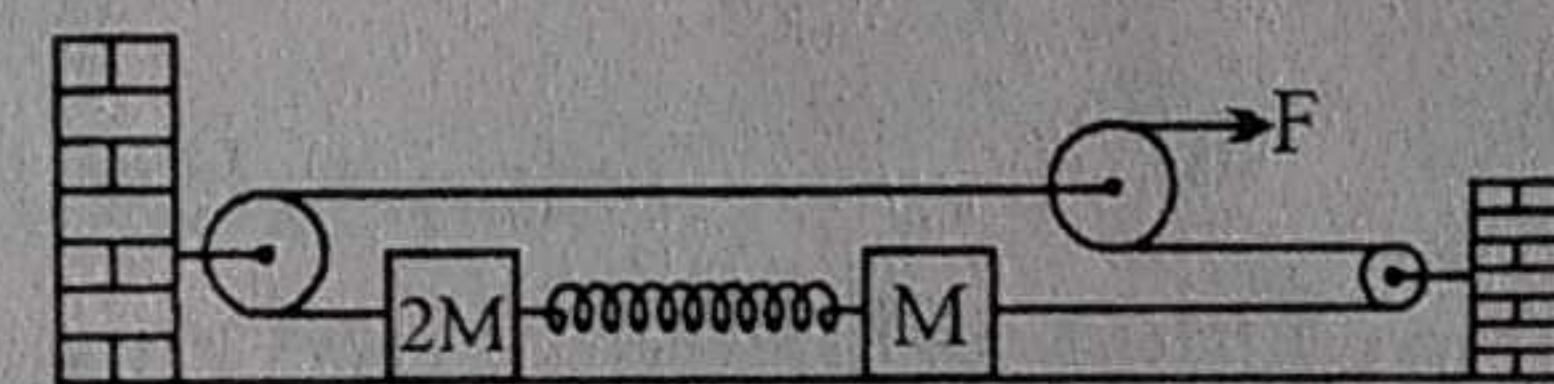
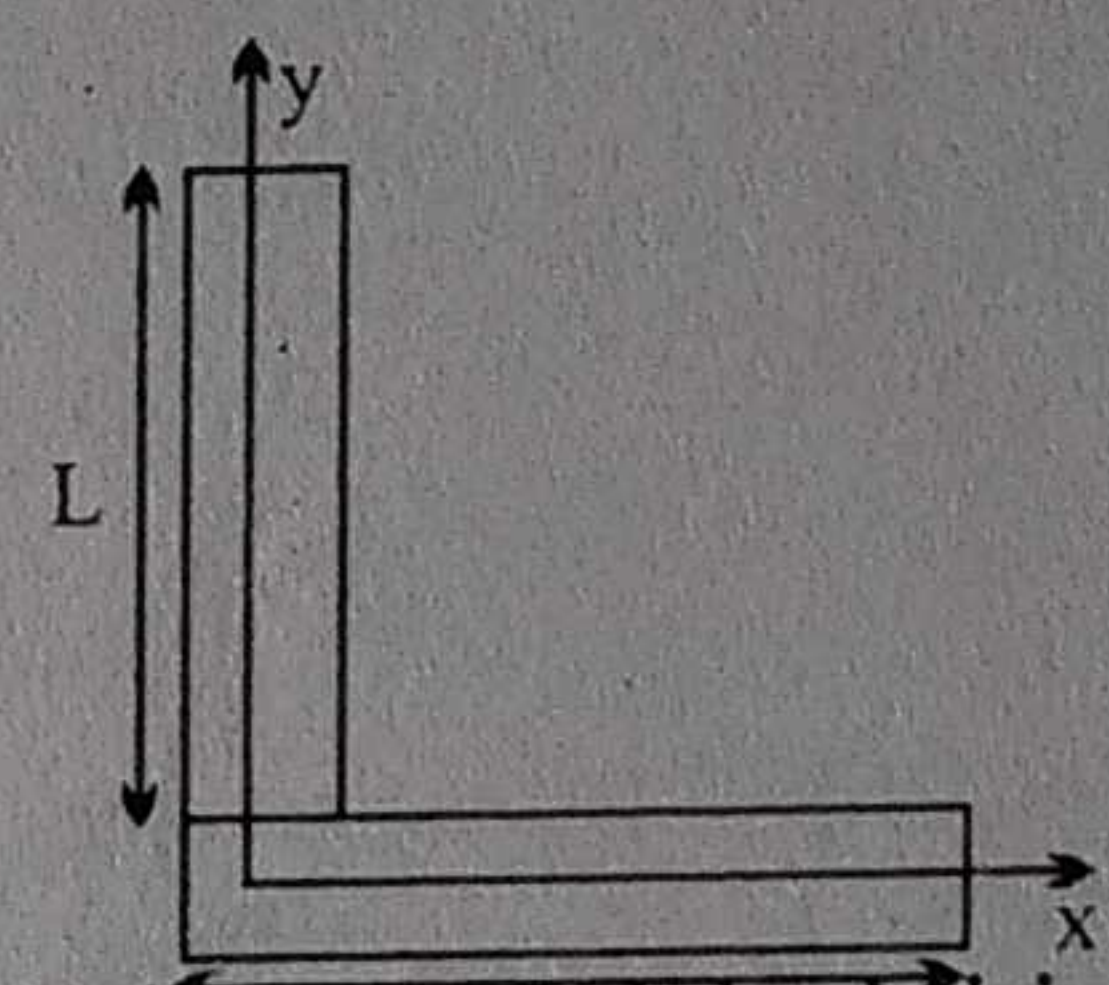
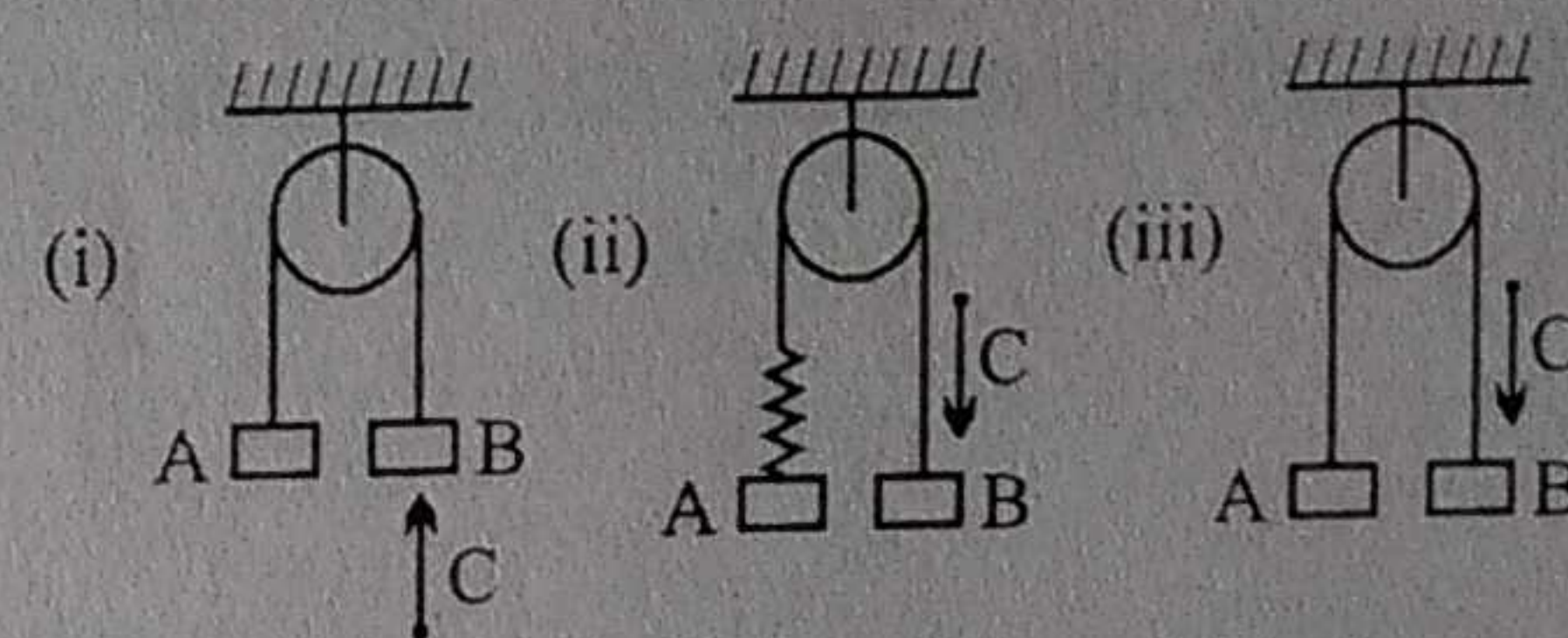


- Q.17 In the figure shown, the two identical balls of mass M and radius R each, are placed in contact with each other on the frictionless horizontal surface. The third ball of mass M and radius $R/2$, is coming down vertically and has a velocity $= v_0$ when it simultaneously hits the two balls and itself comes to rest. Then, each of the two bigger balls will move after collision with a speed equal to

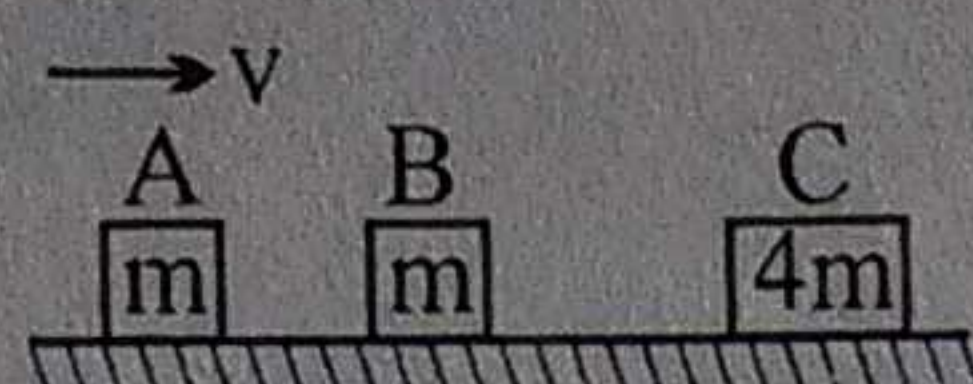
- (A) $4v_0/\sqrt{5}$ (B) $2v_0/\sqrt{5}$ (C) $v_0/\sqrt{5}$ (D) None



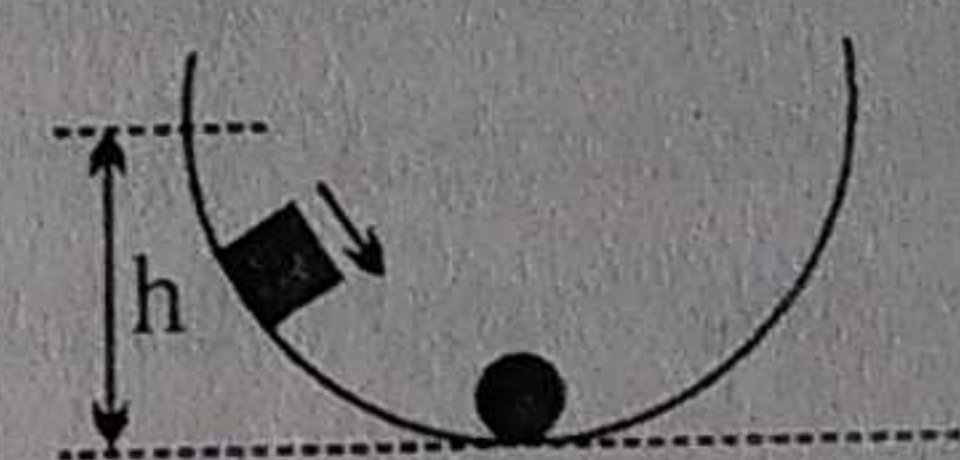
- Q.18 In the **above**, suppose that the smaller ball does not stop after collision, but continues to move downwards with a speed $= v_0/2$, after the collision. Then, the speed of each bigger ball after collision is
 (A) $4v_0/\sqrt{5}$ (B) $2v_0/\sqrt{5}$ (C) $v_0/2\sqrt{5}$ (D) None
- Q.19 A body of mass 'm' is dropped from a height of 'h'. Simultaneously another body of mass 2m is thrown up vertically with such a velocity v that they collide at the height h/2. If the collision is perfectly inelastic, the velocity at the time of collision with the ground will be :
 (A) $\sqrt{\frac{5gh}{4}}$ (B) $\sqrt{gh}$ (C) $\sqrt{\frac{gh}{4}}$ (D) $\frac{\sqrt{10gh}}{3}$
- Q.20 In the figure (i), (ii) & (iii) shown the objects A, B & C are of same mass. String, spring & pulley are massless. C strikes B with velocity 'u' in each case and sticks to it. The ratio of velocity of B in case (i) to (ii) to (iii) is
 (A) 1 : 1 : 1 (B) 3 : 3 : 2
 (C) 3 : 2 : 2 (D) none of these
- Q.21 Centre of mass of two thin uniform rods of same length but made up of different materials & kept as shown, can be, if the meeting point is the origin of co-ordinates
 (A) $(L/2, L/2)$ (B) $(2L/3, L/2)$
 (C) $(L/3, L/3)$ (D) $(L/3, L/6)$
- Q.22 A force exerts an impulse I on a particle changing its speed from u to 2u. The applied force and the initial velocity are oppositely directed along the same line. The work done by the force is
 (A) $\frac{3}{2} I u$ (B) $\frac{1}{2} I u$ (C) $I u$ (D) $2 I u$
- Q.23 In the diagram shown, no friction at any contact surface. Initially, the spring has no deformation. What will be the maximum deformation in the spring? Consider all the strings to be sufficiently large. Consider the spring constant to be K.
 (A) $4F / 3K$ (B) $8F / 3K$
 (C) $F / 3K$ (D) none
- Q.24 In a smooth stationary cart of length d, a small block is projected along its length with velocity v towards front. Coefficient of restitution for each collision is e. The cart rests on a smooth ground and can move freely. The time taken by block to come to rest w.r.t. cart is
 (A) $\frac{ed}{(1-e)v}$ (B) $\frac{ed}{(1+e)v}$ (C) $\frac{d}{e}$ (D) infinite
- Q.25 An open water tight railway wagon of mass 5×10^3 kg coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be
 (A) 0.5 m/s (B) 2 m/s (C) 1 m/s (D) 1.5 m/s
- Q.26 A parallel beam of particles of mass m moving with velocity v impinges on a wall at an angle θ to its normal. The number of particles per unit volume in the beam is n. If the collision of particles with the wall is elastic, then the pressure exerted by this beam on the wall is :
 (A) $2 mn v^2 \cos \theta$ (B) $2 mn v^2 \cos^2 \theta$ (C) $2 mn v \cos \theta$ (D) $2 mn v \cos^2 \theta$
- Q.27 A rocket of mass 4000 kg is set for vertical firing. How much gas must be ejected per second so that the rocket may have initial upwards acceleration of magnitude 19.6 m/s^2 . [Exhaust speed of fuel = 980 m/s .]
 (A) 240 kg s^{-1} (B) 60 kg s^{-1} (C) 120 kg s^{-1} (D) None



- Q.28 Three blocks are initially placed as shown in the figure. Block A has mass m and initial velocity v to the right. Block B with mass m and block C with mass $4m$ are both initially at rest. Neglect friction. All collisions are elastic. The final velocity of block A is

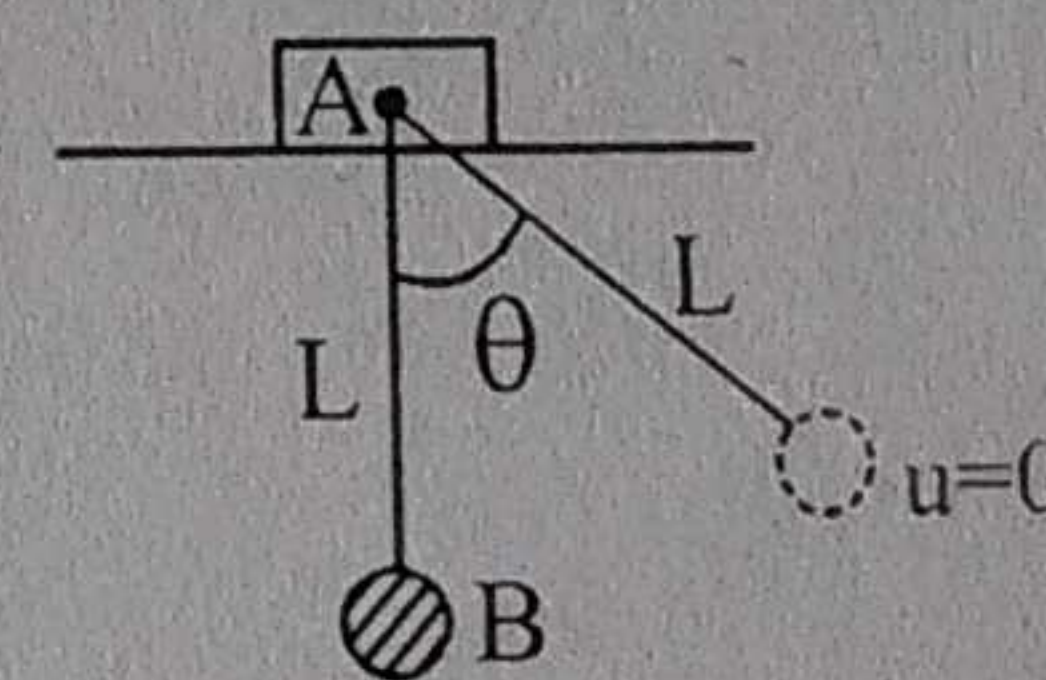


- (A) $0.6v$ to the left
(B) $1.4v$ to the left
(C) v to the left
(D) $0.4v$ to the right
- Q.29 There are some passengers inside a stationary railway compartment. The track is frictionless. The centre of mass of the compartment itself (without the passengers) is C_1 , while the centre of mass of the 'compartment plus passengers' system is C_2 . If the passengers move about inside the compartment along the track.
- (A) both C_1 and C_2 will move with respect to the ground
(B) neither C_1 nor C_2 will move with respect to the ground
(C) C_1 will move but C_2 will be stationary with respect to the ground
(D) C_2 will move but C_1 will be stationary with respect to the ground
- Q.30 A block of mass m starts from rest and slides down a frictionless semi-circular track from a height h as shown. When it reaches the lowest point of the track, it collides with a stationary piece of putty also having mass m . If the block and the putty stick together and continue to slide, the maximum height that the block-putty system could reach is:
- (A) $h/4$
(B) $h/2$
(C) h
(D) independent of h
- Q.31 A boy hits a baseball with a bat and imparts an impulse J to the ball. The boy hits the ball again with the same force, except that the ball and the bat are in contact for twice the amount of time as in the first hit. The new impulse equals:
- (A) half the original impulse
(B) the original impulse
(C) twice the original impulse
(D) four times the original impulse
- Q.32 Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as ball 2. Initially, ball 1 moves with a speed v towards ball 2 which is at rest. Immediately after the collision, ball 1 travels at a speed of $v/3$ in the same direction. What type of collision has occurred?
- (A) inelastic
(B) elastic
(C) completely inelastic
(D) Cannot be determined from the information given



Question No. 105 to 108 (4 questions)

A small ball B of mass m is suspended with light inelastic string of length L from a block A of same mass m which can move on smooth horizontal surface as shown in the figure. The ball is displaced by angle θ from equilibrium position & then released.



- Q.33 The displacement of block when ball reaches the equilibrium position is
- (A) $\frac{L \sin \theta}{2}$
(B) $L \sin \theta$
(C) L
(D) none of these
- Q.34 Tension in string when it is vertical, is
- (A) mg
(B) $mg(2 - \cos \theta)$
(C) $mg(3 - 2 \cos \theta)$
(D) none of these
- Q.35 Maximum velocity of block during subsequent motion of the system after release of ball is
- (A) $[gl(1 - \cos \theta)]^{1/2}$
(B) $[2gl(1 - \cos \theta)]^{1/2}$
(C) $[gl \cos \theta]^{1/2}$
(D) informations are insufficient to decide

Q.36 The displacement of centre of mass of A + B system till the string becomes vertical is
 (A) zero (B) $\frac{L}{2}(1 - \cos \theta)$ (C) $\frac{L}{2}(1 - \sin \theta)$ (D) none of these

Q.37 A system of N particles is free from any external forces.

(a) Which of the following is true for the magnitude of the total momentum of the system?

- (A) It must be zero
 (B) It could be non-zero, but it must be constant
 (C) It could be non-zero, and it might not be constant
 (D) The answer depends on the nature of the internal forces in the system

(b) Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?

- (A) It must be zero
 (B) It could be non-zero, but it must be constant
 (C) It could be non-zero, and it might not be constant
 (D) It could be zero, even if the magnitude of the total momentum is not zero

Q.38 An isolated rail car of mass M is moving along a straight, frictionless track at an initial speed v_0 . The car is passing under a bridge when a crate filled with N bowling balls, each of mass m, is dropped from the bridge into the bed of the rail car. The crate splits open and the bowling balls bounce around inside the rail car, but none of them fall out.

(a) Is the momentum of the rail car + bowling balls system conserved in this collision?

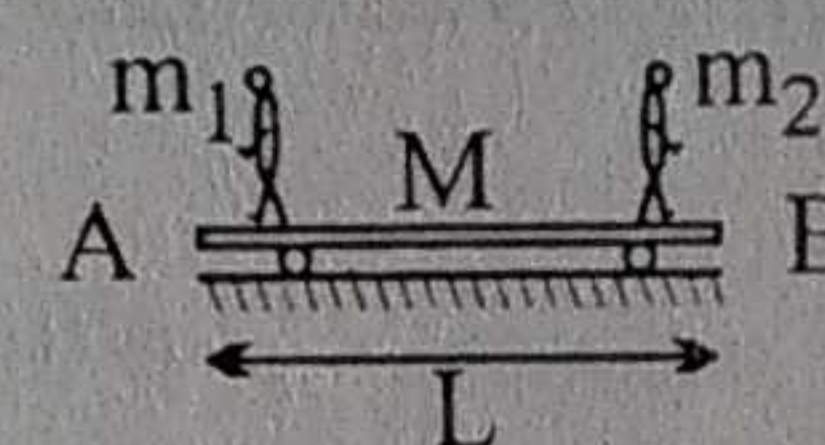
- (A) Yes, the momentum is completely conserved.
 (B) Only the momentum component in the vertical direction is conserved.
 (C) Only the momentum component parallel to the track is conserved.
 (D) No components are conserved.

(b) What is the average speed of the rail car + bowling balls system some time after the collision?

- (A) $(M + Nm)v_0/M$ (B) $Mv_0/(Nm + M)$ (C) Nmv_0/M
 (D) The speed cannot be determined because there is not enough information

Question No. 111 to 115 (5 questions)

Two persons of mass m_1 and m_2 are standing at the two ends A and B respectively, of a trolley of mass M as shown.



Q.39 When the person standing at A jumps from the trolley towards left with u_{rel} with respect to the trolley, then

- (A) the trolley moves towards right
 (B) the trolley rebounds with velocity $\frac{m_1 u_{rel}}{m_1 + m_2 + M}$
 (C) the centre of mass of the system remains stationary
 (D) all the above

Q.40 When only the person standing at B jumps from the trolley towards right while the person at A keeps standing, then

- (A) the trolley moves towards left
 (B) the trolley moves with velocity $\frac{m_2 u_{rel}}{m_1 + m_2 + M}$

- (C) the centre of mass of the system remains stationary
(D) all the above

- Q.41 When both the persons jump simultaneously with same speed then
(A) the centre of mass of the system remains stationary
(B) the trolley remains stationary
(C) the trolley moves toward the end where the person with heavier mass is standing
(D) None of these
- Q.42 When both the persons jump simultaneously with u_{rel} with respect to the trolley, then the velocity of the trolley is
(A) $\frac{|m_1 - m_2| u_{rel}}{m_1 + m_2 + M}$ (B) $\frac{|m_1 - m_2| u_{rel}}{M}$ (C) $\left| \frac{m_1 u_{rel}}{m_2 + M} - \frac{m_2 u_{rel}}{m_1 + M} \right|$ (D) none of these
- Q.43 Choose the incorrect statement, if $m_1 = m_2 = m$ and both the persons jump one by one, then
(A) the centre of mass of the system remains stationary
(B) the final velocity of the trolley is in the direction of the person who jumps first
(C) the final velocity of the trolley is $\left(\frac{mu_{rel}}{M + m} - \frac{mu_{rel}}{M + 2m} \right)$
(D) none of these
- Q.44 A ball is dropped from a height h . As it bounces off the floor, its speed is 80 percent of what it was just before it hit the floor. The ball will then rise to a height of most nearly
(A) 0.80 h (B) 0.75 h (C) 0.64 h (D) 0.50 h
- Q.45 A ball is thrown vertically downwards with velocity $\sqrt{2gh}$ from a height h . After colliding with the ground it just reaches the starting point. Coefficient of restitution is
(A) $1/\sqrt{2}$ (B) $1/2$ (C) 1 (D) $\sqrt{2}$
- Q.46 A ball is dropped from height 5m. The time after which ball stops rebounding if coefficient of restitution between ball and ground $e = 1/2$, is
(A) 1 sec (B) 2 sec (C) 3 sec (D) infinite
- Q.47 A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collision with a vertical wall and returns to ground. The total time of flight of the ball is
(A) $\frac{2v \sin \theta}{g}$ (B) $\frac{2v \cos \theta}{g}$ (C) $\frac{v \sin 2\theta}{g}$ (D) $\frac{v \cos \theta}{g}$

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Take approx. 3 minutes for answering each question.

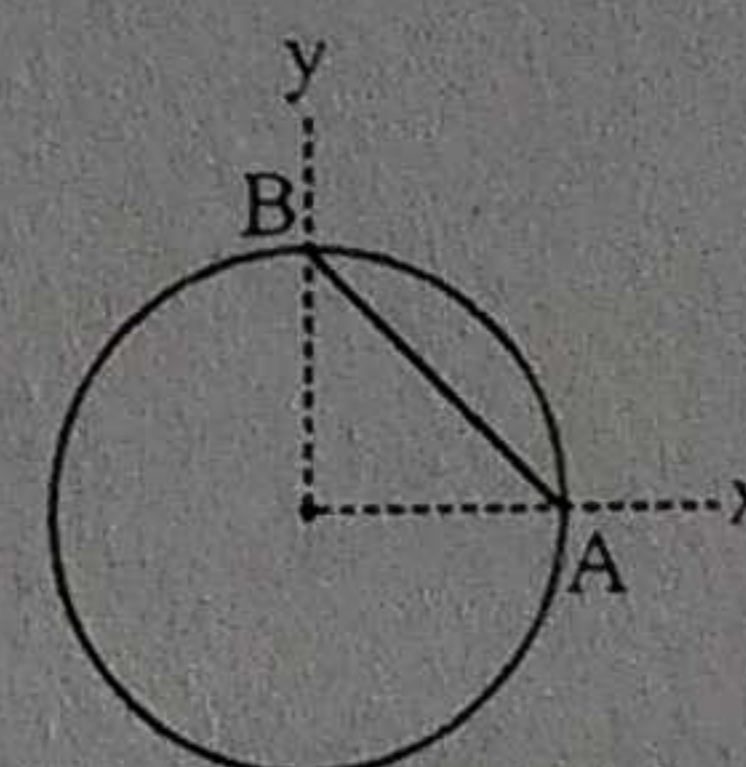
- Q.1 A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle which has twice its mass during the impact.
(A) The minimum kinetic energy of the system is 1 joule.
(B) The maximum elastic potential energy of the system is 2 joule.
(C) Momentum and total kinetic energy of the system are conserved at every instant.
(D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.

Q.2 Two balls A and B having masses 1 kg and 2 kg, moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then correct statements is :

- (A) The velocity of B after collision is 6 m/s opposite to its direction of motion before collision.
 (B) The coefficient of restitution is 0.2.
 (C) The loss of kinetic energy due to collision is 200 J.
 (D) The impulse of the force between the two balls is 40 Ns.

Q.3 An object comprises of a uniform ring of radius R and its uniform chord AB (not necessarily made of the same material) as shown. Which of the following can not be the centre of mass of the object

- (A) $(R/3, R/3)$ (B) $(R/3, R/2)$
 (C) $(R/4, R/4)$ (D) $(R/\sqrt{2}, R/\sqrt{2})$



Q.4 Consider following statements

- [1] CM of a uniform semicircular disc of radius $R = 2R/\pi$ from the centre
 [2] CM of a uniform semicircular ring of radius $R = 4R/3\pi$ from the centre
 [3] CM of a solid hemisphere of radius $R = 4R/3\pi$ from the centre
 [4] CM of a hemisphere shell of radius $R = R/2$ from the centre

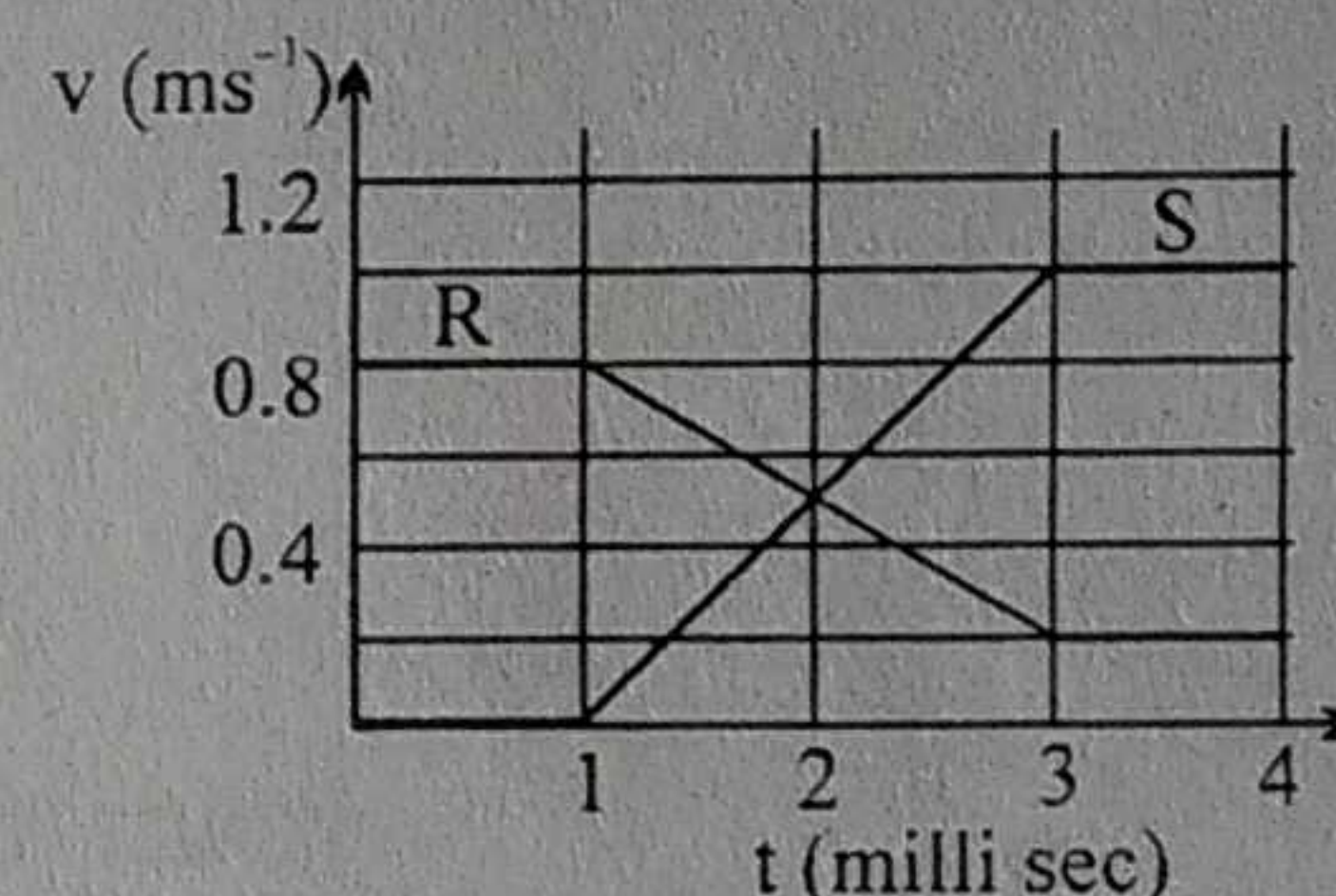
Which statements are correct?

- (A) 1, 2, 4 (B) 1, 3, 4 (C) 4 only (D) 1, 2 only

Q.5 The diagram to the right shows the velocity-time graph for two masses R and S that collided elastically. Which of the following statements is true?

- (I) R and S moved in the same direction after the collision.
 (II) Kinetic energy of the system (R & S) is minimum at $t = 2$ milli sec.
 (III) The mass of R was greater than mass of S.

- (A) I only (B) II only (C) I and II only (D) I, II and III



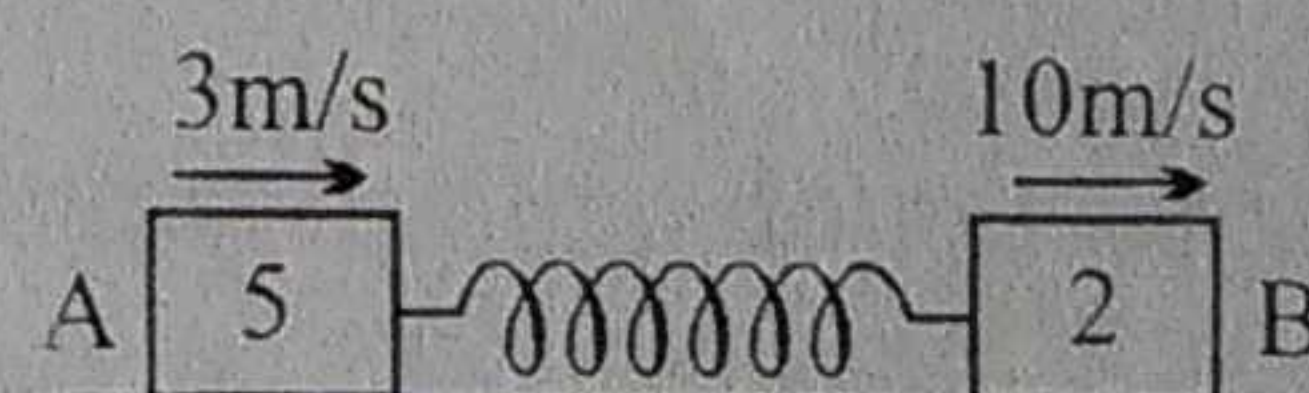
Q.6 In an inelastic collision,

- (A) the velocity of both the particles may be same after the collision
 (B) kinetic energy is not conserved
 (C) linear momentum of the system is conserved.
 (D) velocity of separation will be less than velocity of approach.

Q.7 Two identical balls are interconnected with a massless and inextensible thread. The system is in gravity free space with the thread just taut. Each ball is imparted a velocity v , one towards the other ball and the other perpendicular to the first, at $t = 0$. Then,

- (A) the thread will become taut at $t = (L/v)$
 (B) the thread will become taut at some time $t < (L/v)$.
 (C) the thread will always remain taut for $t > (L/v)$.
 (D) the kinetic energy of the system will always remain mv^2 .

Q.8 Two blocks A (5kg) and B (2kg) attached to the ends of a spring constant 1120 N/m are placed on a smooth horizontal plane with the spring undeformed. Simultaneously velocities of 3 m/s and 10 m/s along the line of the spring in the same direction are imparted to A and B then



(A) when the extension of the spring is maximum the velocities of A and B are zero.

(B) the maximum extension of the spring is 25cm.

(C) maximum extension and maximum compression occur alternately.

(D) the maximum compression occur for the first time after $\frac{\pi}{56}$ sec.

Q.9 In a one-dimensional collision between two particles, their relative velocity is $\vec{v}_1$ before the collision and $\vec{v}_2$ after the collision

(A) $\vec{v}_1 = \vec{v}_2$ if the collision is elastic

(B) $\vec{v}_1 = -\vec{v}_2$ if the collision is elastic

(C) $|\vec{v}_2| = |\vec{v}_1|$ in all cases

(D) $\vec{v}_1 = -k\vec{v}_2$ in all cases, where $k \geq 1$

Q.10 An isolated rail car originally moving with speed v_0 on a straight, frictionless, level track contains a large amount of sand. A release valve on the bottom of the car malfunctions, and sand begins to pour out straight down relative to the rail car.

(a) Is momentum conserved in this process?

(A) The momentum of the rail car alone is conserved

(B) The momentum of the rail car + sand remaining within the car is conserved

(C) The momentum of the rail car + all of the sand, both inside and outside the rail car, is conserved

(D) None of the three previous systems have momentum conservation

(b) What happens to the speed of the rail car as the sand pours out?

(A) The car begins to roll faster

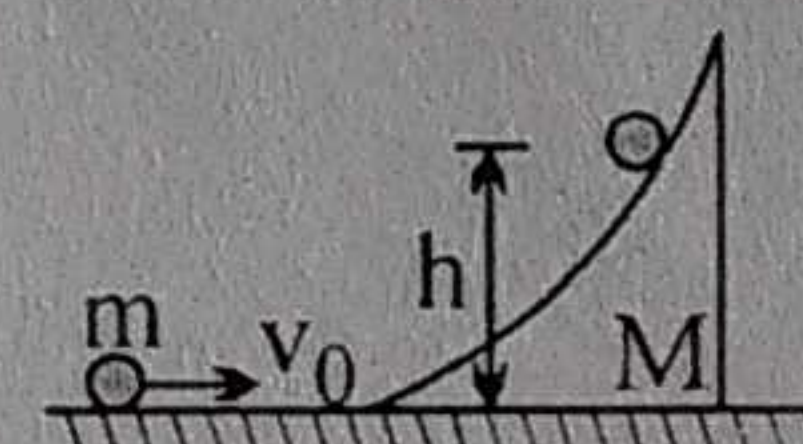
(B) The car maintains the same speed

(C) The car begins to slow down

(D) The problem cannot be solved since momentum is not conserved

Question No. 34 to 40 (7 questions)

A particle of mass m moving horizontally with v_0 strikes a smooth wedge of mass M , as shown in figure. After collision, the ball starts moving up the inclined face of the wedge and rises to a height h .



Q.11 The final velocity of the wedge v_2 is

(A) $\frac{mv_0}{M}$

(B) $\frac{mv_0}{M+m}$

(C) v_0

(D) insufficient data

Q.12 When the particle has risen to a height h on the wedge, then choose the correct alternative(s)

(A) The particle is stationary with respect to ground

(B) Both are stationary with respect to the centre of mass

(C) The kinetic energy of the centre of mass remains constant

(D) The kinetic energy with respect to centre of mass is converted into potential energy

Q.13 The maximum height h attained by the particle is

(A) $\left(\frac{m}{m+M}\right)\frac{v_0^2}{2g}$

(B) $\left(\frac{m}{M}\right)\frac{v_0^2}{2g}$

(C) $\left(\frac{M}{m+M}\right)\frac{v_0^2}{2g}$

(D) none of these

Q.14 Identify the correct statement(s) related to the situation when the particle starts moving downward.

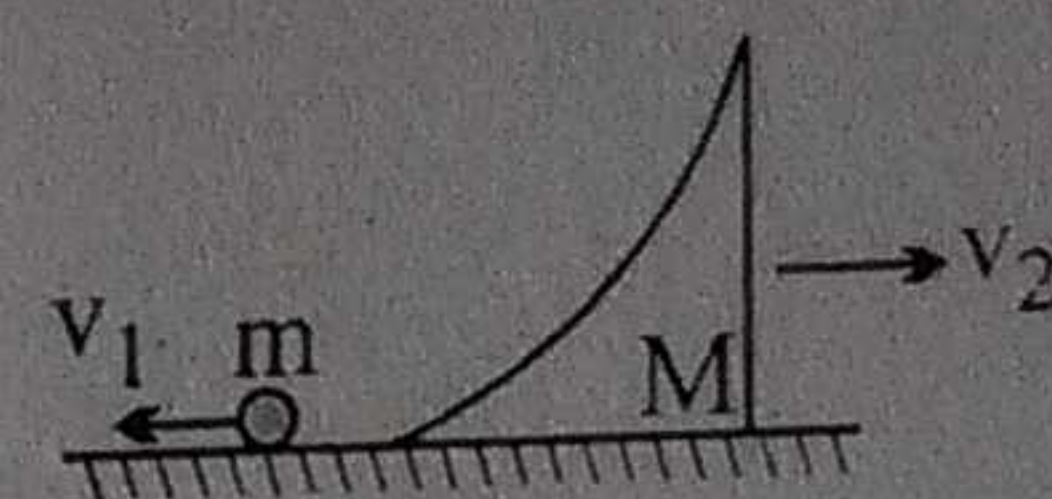
(A) The centre of mass of the system remains stationary

(B) Both the particle and the wedge remain stationary with respect to centre of mass

(C) When the particle reaches the horizontal surface its velocity relative to the wedge is v_0

(D) None of these

Q.15 Suppose the particle when reaches the horizontal surfaces, its velocity with respect to ground is v_1 and that of wedge is v_2 . Choose the correct statement(s)



- (A) $mv_1 = Mv_2$ (B) $Mv_2 - mv_1 = mv_0$ (C) $v_1 + v_2 = v_0$ (D) $v_1 + v_2 < v_0$

Q.16 Choose the correct statement(s) related to particle m

(A) Its kinetic energy is $K_f = \left(\frac{mM}{m+M} \right) gh$

(B) $v_1 = v_0 \left(\frac{M-m}{M+m} \right)$

(C) The ratio of its final kinetic energy to its initial kinetic energy is $\frac{K_f}{K_i} = \left(\frac{M}{m+M} \right)^2$

(D) It moves opposite to its initial direction of motion

Q.17 Choose the correct statement related to the wedge M

(A) Its kinetic energy is $K_f = \left(\frac{4m^2}{m+M} \right) gh$

(B) $v_2 = \left(\frac{2m}{m+M} \right) v_0$

(C) Its gain in kinetic energy is $\Delta K = \left(\frac{4mM}{(m+M)^2} \right) \left(\frac{1}{2} mv_0^2 \right)$

(D) Its velocity is more than the velocity of centre of mass

ANSWER KEY

ONLY ONE OPTION IS CORRECT.

Q.1	B	Q.2	B	Q.3	B	Q.4	D	Q.5	B	Q.6	B	Q.7	C
Q.8	C	Q.9	C	Q.10	A	Q.11	A	Q.12	C	Q.13	D	Q.14	D
Q.15	A	Q.16	B	Q.17	C	Q.18	C	Q.19	D	Q.20	B	Q.21	D
Q.22	B	Q.23	B	Q.24	D	Q.25	C	Q.26	B	Q.27	C	Q.28	A
Q.29	C	Q.30	A	Q.31	C	Q.32	B	Q.33	A	Q.34	D	Q.35	A
Q.36	B	Q.37(a) B (b) C				Q.38(a) C (b) B				Q.39	D	Q.40	D
Q.41	A	Q.42	A	Q.43	D	Q.44	C	Q.45	A	Q.46	C	Q.47	B

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

Q.1	A,B,D	Q.2	A,B,C	Q.3	B,D	Q.4	C	Q.5	D	Q.6	A,B,C,D
Q.7	A,C	Q.8	B,C	Q.9	B,D	Q.10(a) A,C (b) B				Q.11	B
Q.12	B,D	Q.13	C	Q.14	C	Q.15	B,C	Q.16	B	Q.17	A,B,C,D